



Unit Outline

COS20028

Big Data Architecture and Application

Semester May 2025

Please read this Unit Outline carefully. It includes:

PART A Unit summary

PART B Your Unit in more detail

PART C Further information



"Swinburne University of Technology recognises the historical and cultural significance of Australia's Indigenous history and the role it plays in contemporary education

Each day in Australia, we all walk on traditional Indigenous land

We therefore acknowledge the traditional custodians of the land that our Australian campuses currently occupy, the Wurundjeri people, and pay respect to Elders past and present, including those from other areas who now reside on Wurundjeri land"

PART A: Unit Summary

Unit Code(s)		COS20028
Unit Title		Big Data Architecture and Application
Duration		One Semester
Total Contact Hours		48 hours
Requisites:		
	Pre-requisites	[COS10022 – Introduction to Data Science] AND [COS20007 – Object-Oriented Programming or COS30016 – Programming in Java]
	Co-requisites	Nil
	Concurrent pre requisites	Nil
	Anti-requisites	Nil
	Assumed knowledge	Object-oriented programming skills are expected.
Credit Points		12.5 credit points
Campus/Location		Ha Noi Location
Mode of Delivery		Blended
Assessment Summary		Assignment 1 (Individual, 20%) Assignment 2 (Individual, 30%) Online Test 1 (Individual, 10%) Online Test 2 (Individual, 10%) Online Test 3 (Individual, 30%)

Aims

This unit introduces students to the architectures of contemporary Big Data platforms and ecosystems of tools. These technologies are the foundation of Big Data analytics, which facilitate scalable management and processing of vast quantities of data. The unit is delivered in collaboration with industry and prepares students with the essential foundations towards undertaking professional certifications.

Unit Learning Outcomes

Students who successfully complete this unit can:

1. Explain and compare the architecture of contemporary Big Data tools and platforms.
2. Demonstrate skills in working with a contemporary Big Data platform to manage large data sets.
3. Apply Big Data related technologies in developing applications to solve common problems faced by organisations.

Graduate Attributes

The Swinburne Graduate Attributes describe the capability of our graduates to use knowledge, skills and behaviours to contribute to society meaningfully and positively. They include professional, self-directed learning and future-ready skills.

This unit contributes to the development of the following Swinburne Graduate Attributes:

- GA1 Communication - Verbal communication: Assignment 1 requires a video recording to explain the design idea of your solution.
- GA2 Communication - Communicating using different media: Assignment 2 requires multiple media and tools to complete the content.
- GA5 Digital literacies– Information literacy: All assignments and tests require students to process the data with the knowledge of decomposing the structure in the data and categorising them by the extracted information.
- GA6 Digital Literacies– Technical literacy: Assignment 2, online test 2, and online test 3 require students to be familiar with different tools to be used for corresponding purposes for big data.

Other graduate attributes may be practised in the unit but are not formally taught as part of the unit content, nor incorporated within formal assessment.

Content

- Introduction to Big Data platforms and their ecosystems
- Big Data platform programming
- Partitioners and Reducers
- Big Data acquisition and workflows
- The Data Analytics Lifecycle
- Analysis and Management of structured and unstructured Big Data ▪

Graduate Attribute – Communication Skills: Verbal communication

- Graduate Attribute – Communication Skills: Communicating using different media
- Graduate Attribute – Digital Literacies: Information literacy
- Graduate Attribute – Digital Literacies: Technical literacy

PART B: Your Unit in more detail

Unit Improvements

Feedback provided by previous students through the Student Survey has resulted in improvements that have been made to this unit. Recent improvements include:

- The VM is rebuilt with updated software and includes the new tool.
- New subject is included to replace the obsolete subject.
- Assignment activities are updated.

Unit Teaching Staff

Name	Role	Email	Consultation Times
Dr. Pham Thi Kim Dung (Doris)	Unit Convenor	dtpham@swin.edu.au	By email appointment

Learning and Teaching Structure

Category	Activity	Total Hours	Hours per Week	Teaching Period Weeks
On Campus	Lectures	24 hours	2 hours	Weeks 1 to 12
On Campus	Tutorials	24 hours	2 hours	Weeks 1 to 12
Unspecified Activities	Independent Learning (Directed Online Learning and Independent Learning)	102 hours	8.5 hours	Weeks 1 to 12

Week by Week Schedule

Week	Week Beginning	Teaching and Learning Activity	Student Task or Assessment
1	May 05	▪ Lecture: Overview of Hadoop. ▪ Online materials: Week 1 lecture slides and videos. Lab: Setting up and configuring the VM environment. Getting familiar with basic HDFS operation commands.	▪ Readings: Chapter 1 of "Hadoop: The Definitive Guide, 4th Edition." ▪ Set up and access Hadoop environment. ▪ Using HDFS.
2	May 12	▪ Lecture: Hadoop Ecosystem: components, clusters, MapReduce. ▪ Online materials: Week 2 lecture slides and videos.	▪ Readings: Chapter 2 of "Hadoop: The Definitive Guide, 4th Edition." Running a MapReduce job.

		Lab: Executing and understanding a MapReduce job.	
3	May 19	▪ Lecture: MapReduce Programming I: Developing the first MapReduce program with Java and Streaming API (Python). ▪ Online materials: Week 3 lecture slides and videos. Lab: Work on LT1 contents and the online test 1.	▪ Readings: Chapter 5 of “Hadoop: The Definitive Guide, 4th Edition.” ▪ Using ToolRunner and passing parameters. ▪ Using a Combiner. Online Test 1. (Covering Week 1 – 3 materials.)
4	May 26	▪ Lecture: MapReduce Programming II: Hadoop APIs. ▪ Online materials: Week 4 lecture slides and videos. Lab: ToolRunner, Combiner, LT2 contents.	▪ Readings: Chapter 6 of “Hadoop: The Definitive Guide, 4th Edition.” ▪ Testing with LocalJobRunner. ▪ Logging. ▪ ToolRunner. ▪ Combiner.
5	Jun 02	▪ Lecture: MapReduce Programming III: Tips and Techniques (Debugging, LocalJobRunner, Log Files, Counters, Object Reuse, Map-Only Jobs). ▪ Online materials: Week 5 lecture slides and videos. Lab: Work on LocalJobRunner, Logging, and LT3 contents.	▪ Readings: Chapter 4 of “Hadoop: The Definitive Guide, 4th Edition.” ▪ Writing a Partitioner. ▪ LocalJobRunner ▪ Logging. Assignment 1 Due.
6	Jun 09	▪ Lecture: MapReduce Programming IV: Partitioners and reducers, Data Input and Output. ▪ Online materials: Week 6 lecture slides and videos. Lab: Work on Partitioner, Custom WritableComparable, SequenceFile, File Compression, and the online test 2.	▪ Readings: Chapter 3 of “Hadoop: The Definitive Guide, 4th Edition.” ▪ Creating an Inverted Index ▪ Calculating Word Cooccurrence. ▪ Implementing a custom WritableComparable. ▪ Using Sequence Files and File Compression. Online Test 2. (Covering Week 4 – 6 materials.)
7	Jun 16	▪ Lecture: MapReduce Programming V: Common MapReduce algorithm. Introduction to Kafka. ▪ Online materials: Week 7 lecture slides and videos. Lab: Work on Inverted Index,	▪ Readings: Chapter 8 of “Hadoop: The Definitive Guide, 4th Edition.” ▪ Importing data with Sqoop. ▪ Running an Oozie workflow. Exploring a Secondary Sort example.

		and Word Co-occurrence	
8	Jun 23	▪ Lecture: Data Processing and Analysis I: Sqoop, Flume, Oozie, Kafka. ▪ Online materials: Week 8 lecture slides and videos.	▪ Readings: Chapter 15 of “Hadoop: The Definitive Guide, 4th Edition.” ▪ Data ingest with Hadoop tools. ▪ Exploring a Secondary Sort Example. ▪ Importing Data with Sqoop. ▪ Kafka exercise.
		Lab: Secondary sort, Sqoop data import, and Kafka application.	
9	Jun 30	▪ Lecture: Data Processing and Analysis II: Pig Latin and ETL. ▪ Online materials: Week 9 lecture slides and videos. Lab: Data ingestion, Pig for ETL, Analysing AD campaign, and LT5 contents.	▪ Readings: Chapter 16 of “Hadoop: The Definitive Guide, 4th Edition.” ▪ Analysing disparate datasets with Pig. ▪ Analysing Ad campaign data with Pig.
		Summer Holiday. No classes from 07 – 13 July, inclusively.	
10	Jul 14	▪ Lecture: Data Processing and Analysis III: Pig. ▪ Online materials: Week 10 lecture slides and videos. Lab: Analysing disparate data with Pig and LT6.	▪ Readings: Chapter 17 of “Hadoop: The Definitive Guide, 4th Edition.” ▪ Pig Latin for analysing disparate data.
11	Jul 21	▪ Lecture: Data Processing and Analysis IV: Hive. ▪ Online materials: Week 11 lecture slides and videos. Lab: HiveQL and LT7 content.	▪ Readings: Readings: Chapter 12 of “Hadoop: The Definitive Guide, 4th Edition.” ▪ Sentiment analysis with Hive. ▪ Data transformation with Hive. ▪ Alter and drop a table. ▪ Assignment 2 Due.
12	Jul 28	▪ Lecture: Contemporary Issues in Big Data Analytics. ▪ Online materials: Articles on Blogs. Lab: Online test 3.	▪ Readings: Readings: Chapter 22 of “Hadoop: The Definitive Guide, 4th Edition.” Online test 3. (Covering week 7 – 12 materials.)

Assessment

a) Assessment Overview

Tasks and Details	Individual or Group	Weighting	Mapped Unit Learning Outcomes	Mapped Graduate Attributes	Assessment Due Date
Assignment 1	Individual	20%	2	1, 5	Week 05
Assignment 2	Individual	30%	1, 2, 3	2, 5, 6	Week 11
Online Test 1	Individual	10%	1, 2, 3	5	Week 03 Class
Online Test 2	Individual	10%	1, 2, 3	5, 6	Week 06 Class
Online Test 3	Individual	30%	1, 2, 3	5, 6	Week 12 Class

b) Minimum requirements to pass this unit

To pass this unit, you must:

- achieve an **overall mark** for the unit of 50% or more,
- submit all assignments, and
- participate in all online tests.

Students who do not participate in all assignments and tests will receive a maximum of 44% as the total mark for the unit.

c) Examinations

The online tests are hosted during class time. You will be expected to be available for the time slots stated in your timetable.

d) Use of generative AI (genAI) in this Unit

The valid use of genAI in this unit is as follows:

- All online tests in this unit should be completed entirely without genAI assistance.
- Assessment 1 and 2: genAI may be used only for learning how to create video and guidance on what to be considered when making the video for the assignments. Any use of genAI must be acknowledged, with prompts and outputs included in an appendix.

e) Submission Requirements

Assignments and other assessments are generally submitted online through the Canvas assessment submission system which integrates with the Turnitin plagiarism checking service.

Please ensure you keep a copy of all assessments that are submitted.

In cases where a hard copy submission is required an Assessment Cover Sheet must be submitted with your assignment. The standard Assessment Cover Sheet is available from the [Submitting work](#) webpage or www.swinburne.edu.au/studentforms/

All submissions must be reviewed by at least one colleagues assigned on Canvas. The peer review should be completed before the assignment submission deadline. A submission is considered completed only if it has been reviewed by peers before the submission deadline.

f) Extensions and Late Submission

Late Submissions - Unless an extension has been approved, late submissions will result in a penalty. You will be penalised 10% of your achieved mark for each working day the task is late, up to a maximum of 5 working days. After 5 working days, a zero result will be recorded.

g) Referencing

To avoid breaching academic integrity, you are required to provide references whenever you include information from other sources in your work and acknowledge when you have used Artificial Intelligence (AI) tools (such as ChatGPT). Further details regarding academic integrity are available in Section C of this document.

Referencing conventions required for this unit is **Swinburne Harvard style**.

Helpful information on referencing can be found at

<http://www.swinburne.edu.au/library/referencing/>

h) Groupwork Guidelines

A group assignment is the collective responsibility of the entire group, and if one member is temporarily unable to contribute, the group should be able to reallocate responsibilities to keep to schedule. In the event of longer-term illness or other serious problems involving a member of group, it is the responsibility of the other members to notify immediately the Unit Convenor or relevant tutor.

Group submissions must be submitted with an Assignment Cover Sheet, signed by all members of the group.

All group members must be satisfied that the work has been correctly submitted. Any penalties for late submission will generally apply to all group members, not just the person who submitted.

Required The required textbook(s) are available from Swinburne Bookshop and Swinburne Library, which is available for students to read online: <http://bookshop.swin.edu.au> , <https://www.swinburne.edu.au/library/search/>

Required Books:

- [Hadoop: the definitive guide](#), White, Tom (Tom E.) author., 4th ed., Sebastopol, California: O'Reilly, 2015.
- [Hadoop operations](#), Sammer, Eric., Loukides, Michael Kosta.; Nash, Courtney.; Romano, Robert (Illustrator), illustrator., First edition., Sebastopol, CA: O'Reilly, 2012

Recommended Reading Materials

The Library has a large collection of resource materials, both texts and current journals. Listed below are some references that will provide valuable supplementary information to this unit. It is also recommended that you explore other sources to broaden your understanding.

[Programming pig: dataflow scripting with hadoop](#), Gates, Alan, author.; Dai, Daniel, 2nd edition., Beijing, China: O'Reilly, 2017The required textbook(s) are available from Swinburne Bookshop and Swinburne Library, which is available for students to read online: <http://bookshop.swin.edu.au> , <https://www.swinburne.edu.au/library/search/>

Required Books:

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- [Hadoop operations](#), Sammer, Eric., Loukides, Michael Kosta.; Nash, Courtney.; Romano, Robert (Illustrator), illustrator., First edition., Sebastopol, CA: O'Reilly, 2012

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[Programming pig: dataflow scripting with hadoop](#), Gates, Alan, author.; Dai, Daniel, 2nd edition., Beijing, China: O'Reilly, 2017

PART C: FURTHER INFORMATION



For further information on any of these topics, refer to Swinburne's Student webpage <http://www.swinburne.edu.au/student/>

Student behaviour and wellbeing

All students are expected to: act with integrity, honesty and fairness; be inclusive, ethical and respectful of others; and appropriately use University resources, information, equipment and facilities. All students are expected to contribute to creating a work and study environment that is safe and free from bullying, violence, discrimination, sexual harassment, vilification and other forms of unacceptable behaviour.

The [Student Charter](#) describes what students can reasonably expect from Swinburne in order to enjoy a quality learning experience. The Charter also sets out what is expected of students with regards to your studies and the way you conduct yourself towards other people and property.

You are expected to familiarise yourself with University regulations and policies and are obliged to abide by these, including the [Student Academic Misconduct Regulations](#), [Student General Misconduct Regulations](#) and the [People, Culture and Integrity Policy](#). Any student found to be in breach of these may be subject to disciplinary processes.

Examples of expected behaviours are:

- conducting yourself in teaching areas in a manner that is professional and not disruptive to others
- following specific safety procedures in Swinburne laboratories, such as wearing appropriate footwear and safety equipment, not acting in a manner which is dangerous or disruptive (e.g. playing computer games), and not bringing in food or drink
- following emergency and evacuation procedures and following instructions given by staff/wardens in an emergency response

Canvas

You should regularly log on to the Swinburne learning management system, Canvas. You can access Canvas via the [Student login](#) webpage or <https://swinburne.instructure.com/>

Canvas is updated regularly with important unit information and communications.

Communication

All communication will be via your Swinburne email address. If you access your email through a provider other than Swinburne, then it is your responsibility to ensure that your Swinburne email is redirected to your private email address.

Academic Integrity

Academic integrity is about taking responsibility for your learning and submitting work that is honestly your own. It means acknowledging the ideas, contributions and work of others; referencing your sources and acknowledging the use of generative artificial intelligence; contributing fairly to group work; and completing tasks, tests and exams without cheating. Artificial intelligence tools should only be used where approved by the Unit Convenor.

Swinburne University uses the Turnitin system, which helps to identify inadequate citations, poor paraphrasing and unoriginal work in assignments that are submitted via Canvas. Your Unit Convenor will provide further details.

Plagiarising, cheating and seeking an unfair advantage in a test, exam or assessment task are all breaches of academic integrity and treated as academic misconduct. Examples of breaches of academic integrity include:

- using the whole or part of computer program written by another person as your own without appropriate acknowledgement
- copying the whole or part of somebody else's work in an assessment, including material from a published work, a website or database, a set of lecture notes, current or past student's work, or any other person's work
- using output from artificial intelligence tools (e.g. ChatGPT) in whole or part without acknowledgement and/or without the approval of the Unit Convenor
- poorly paraphrasing somebody else's work
- using a musical composition or audio, visual, graphic and photographic work created by another without acknowledgment
- using objects, artefacts, costumes or models created by another person and presenting them as your own
- submitting assessments that have been developed by another person or service (paid or unpaid), referred to as contract cheating
- presenting or submitting assignments or other work in conjunction with another person or group of people when that work should be your own independent work.
- enabling others to cheat, including letting another student copy your work or by giving access to a draft or completed assignment.

The penalties for academic misconduct can be severe, ranging from a zero grade for an assessment task through to exclusion from Swinburne.

For further details, see <https://www.swinburne.edu.au/student-login/academic-integrity/>

Student support

Swinburne offers a range of services and resources to help you complete your studies successfully. Your Unit Convenor or studentHQ can provide information about the study support and other services available for Swinburne students. See <https://portal.swin.edu.vn/> for further information.

Special consideration

If your studies have been adversely affected due to serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you may be able to apply for special consideration (SPC).

Applications for Special Consideration are submitted via the SPC online tool normally no later than 5.00pm on the third working day after the submission/sitting date for the relevant assessment component. See

<https://www.swinburne.edu.au/life-at-swinburne/student-support/services/special-consideration-assistance/>

Accessibility needs

Sometimes students with a disability, a mental health or medical condition or significant carer responsibilities require reasonable adjustments to fully access and participate in education. Swinburne's AccessAbility Services can develop an 'Education Access Plan' that includes the

services and reasonable adjustments that you need. The plan makes recommendations to University teaching and examination staff.

It is recommended that you register with AccessAbility Services within one week after the commencement of your unit to allow the University to make reasonable adjustments.

Review of marks

An independent marker reviews all fail grades for major assessment tasks. In addition, a review of assessment is undertaken if your final result is between 45 and 49 or within 2 marks of any grade threshold.

You can ask the Unit Convenor to check the result for an assessment item or your final result. Your request must be made in writing within 10 working days of receiving the result. The Unit Convenor can discuss the marking criteria with you and check the aggregate marks of assessment components to identify if an error has been made. This is known as local resolution.

If you are dissatisfied with the outcome of the local resolution, you can lodge a formal complaint.

Feedback, complaints and suggestions

In the first instance, discuss any issues with your Unit Convenor. If your concerns are not resolved or you would prefer not to deal with your Unit Convenor, then you can complete a feedback form.

See <https://www.swinburne.edu.au/corporate/feedback/>

Advocacy

Should you require assistance with any academic issues, University statutes, regulations, policies and procedures, you are advised to seek advice from an Academic Student Support Officer at Swinburne Student Life.

For an appointment, please call 19006412 or email swin@fe.edu.vn.

For more information, please see <https://portal.swin.edu.vn>